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Global Linear Instability

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inhomogeneous flows in complex domains, BiGlobal and TriGlobal instability, two-dimensional and three-dimensional eigenvalue and initial value problem, modal and nonmodal instability, flow control

Abstract

This article reviews linear instability analysis of flows over or through complex two-dimensional (2D) and 3D geometries. In the three decades since it first appeared in the literature, global instability analysis, based on the solution of the multidimensional eigenvalue and/or initial value problem, is continuously broadening both in scope and in depth. To date it has dealt successfully with a wide range of applications arising in aerospace engineering, physiological flows, food processing, and nuclear-reactor safety. In recent years, nonmodal analysis has complemented the more traditional modal approach and increased knowledge of flow instability physics. Recent highlights delivered by the application of either modal or nonmodal global analysis are briefly discussed. A conscious effort is made to demystify both the tools currently utilized and the jargon employed to describe them, demonstrating the simplicity of the analysis. Hopefully this will provide new impulses for the creation of next-generation algorithms capable of coping with the main open research areas in which step-change progress can be expected by the application of the theory: instability analysis of fully inhomogeneous, 3D flows and control thereof.

1. INTRODUCTION

1.1. Prologue

Instability and laminar-turbulent flow transition are customarily associated with the so-called parallel shear flows (Lin 1955, Drazin & Reid 1981) on which the vast majority of research efforts has been focused over the past century. The modal, eigenvalue problem (EVP) concept has accompanied research in this area throughout, although, starting in the early 1990s, linear stability theory of parallel flows has been extended to include solutions of the initial-value problem (IVP) associated with nonmodal perturbation development (Schmid & Henningson 2001). Reviews of modal parallel flow instability may be found, e.g., in AGARD Report 709 (Advis. Group Aerosp. Res. Dev. Neuilly-Sur-Seine 1984), Kleiser & Zang (1991), Saric et al. (2003), and references therein. Herbert (1988) discussed a secondary instability of parallel and weakly nonparallel flows in the framework of Floquet theory, the latter applicable when amplified primary linear instabilities become sufficiently large so as to modify periodically the underlying one-dimensional (1D) basic state upon which they have developed. Weakly nonparallel shear flow instability has also been reviewed by Huerre & Monkewitz (1990) and Herbert (1997) in two different contexts, outlined below. The present review deals with the primary linear instability of essentially nonparallel, 2D and 3D flows.

Three decades have passed since the first global linear instability publication appeared in the literature (Pierrehumbert & Widnall 1982). Global analysis work in the past century has been reviewed in an earlier paper (Theofilis 2003), which newcomers to the area may still find of interest. Riding on the crest of ever-increasing computing power and a wider availability of open-source libraries for large-scale linear algebra computations, the scope of global instability research has broadened substantially in the past decade, warranting a new look at the subject.

We begin by discussing numerical methods that have enabled linear modal and nonmodal instability analysis in the past 30 years, clarifying the concepts utilized and emphasizing the order of appearance of particular contributions to the literature, aiming at raising awareness of and giving proper credit to seminal works found in early global instability analysis literature. A case in this point is the citation map of the little-known works of Eriksson & Rizzi (1985) and Chiba (1998), and to a lesser extent that of Tezuka & Suzuki (2006), each of which has introduced step-changing technologies in the area of global instability analysis, although hardly any of these contributions is referred to in the (abundant) recent literature.

From the viewpoint of instability physics, we discuss work that appeared in the literature after Theofilis (2003). The key selection criterion is the generation of conclusive new knowledge of flow physics, if possible independently obtained by more than one research group and cross-validated by experiments or full 3D simulations. The intended audience is the fluid mechanics community at large, with the objective of raising awareness of the theory's potential; to the extent possible, unnecessary colloquialism pertinent to the community of experts in the field has been avoided. Readers interested in learning more about nonmodal instability and control are referred to recent articles by Schmid (2007) and Kim & Bewley (2007), respectively. The review by Chomaz (2005) on weakly nonparallel flows is complementary reading to the material presented herein, especially with regard to non-normality and nonlinearity. The very existence of four review articles in this journal within a space of six years testifies to the vigorous developments that the areas of instability and control of nonparallel flows are presently experiencing.

1.2. Definitions

The development in time and space of small-amplitude perturbations superposed upon a given flow can be described by the linearized Navier-Stokes, continuity, and energy equations. In general,

it is not necessary to invoke either a parallel or a weakly nonparallel flow assumption, and hence the flow analyzed with respect to its stability may be any 2D or 3D solution of the equations of motion. Linearization of the latter around that solution follows; primarily steady or unsteady laminar basic flows, $\bar{\mathbf{q}} = (\bar{\rho}, \bar{u}, \bar{v}, \bar{w}, \bar{T})^T$, are considered here. Such basic flows may be provided analytically in few cases and are typically obtained by 2D or 3D direct numerical simulations, potentially exploiting spatial invariances. Steady laminar flows exist only at low Reynolds numbers, but numerical procedures are in place for the recovery of basic flows also at conditions for which linear global instability would be expected, based on continuation (Keller 1977) and selective frequency damping (Åkervik et al. 2006). Recently, global instability analysis has been extended to the area of industrially relevant turbulent mean flow analysis, results of which are reviewed below.

In using the term small-amplitude perturbations, solutions to the IVP

$$\mathcal{B}(\bar{\mathbf{q}}; Re; Ma) \frac{\partial \hat{\mathbf{q}}}{\partial t} = \mathcal{A}(\bar{\mathbf{q}}; Re; Ma) \hat{\mathbf{q}} \quad (1)$$

are denoted, where $\hat{\mathbf{q}}(\mathbf{x}; t) = (\hat{\rho}, \hat{u}, \hat{v}, \hat{w}, \hat{T})^T$ is the vector comprising the amplitude functions of linear density, velocity component, and temperature or pressure perturbations, which are in general inhomogeneous functions of all three spatial coordinates, $\mathbf{x}$, and time, t . The operators $\mathcal{A}$ and $\mathcal{B}$ are associated with the spatial discretization of the linearized continuity, Navier-Stokes, and energy equations of motion and comprise the basic state, $\bar{\mathbf{q}}(\mathbf{x}, t)$, and its spatial derivatives; when turbulent mean flows are analyzed, Equation 1 is extended in line with the turbulence model utilized. In the particular case of 2D basic flows, Equation 1 may be understood as describing the evolution of the complex 2D amplitude functions into which the 3D small-amplitude perturbations may be decomposed, using a Fourier ansatz along the (single) homogeneous spatial direction.

In both 2D and 3D basic flows, Equation 1 may be rewritten¹ as

$$\frac{d\hat{\mathbf{q}}}{dt} = \mathbf{C}\hat{\mathbf{q}}, \quad (2)$$

with $\mathbf{C} = \mathbf{B}^{-1}\mathbf{A}$. In case of steady basic flows, the separability between time and space coordinates in Equation 2 permits the introduction of a Fourier decomposition in time, leading to the generalized matrix EVP

$$\mathbf{A}\hat{\mathbf{q}} = \omega \mathbf{B}\hat{\mathbf{q}}, \quad (3)$$

in which matrices $\mathbf{A}$ and $\mathbf{B}$ discretize the operators $\mathcal{A}$ and $\mathcal{B}$, respectively, and incorporate the boundary conditions. Alternatively, without reference to the separability property, the autonomous system given in Equation 2 has the explicit solution

$$\hat{\mathbf{q}}(t) = e^{\mathbf{C}t} \hat{\mathbf{q}}(0) \equiv \Phi(t) \hat{\mathbf{q}}(0). \quad (4)$$

Here $\hat{\mathbf{q}}(0) \equiv \hat{\mathbf{q}}(t=0)$, and the matrix exponential, $\Phi(t) \equiv e^{\mathbf{C}t}$, is known as the propagator operator (Farrell & Ioannou 1996). A solution of the IVP given in Equation 2 distinguishes between the limits $t \rightarrow 0$ and $t \rightarrow \infty$. Whereas the latter limit may be described by the EVP given in Equation 3, the growth σ of an initial linear perturbation, $\hat{\mathbf{q}}(0)$, may be computed at all times via

$$\sigma^2 = \frac{\langle e^{\mathbf{C}^*t} e^{\mathbf{C}t} \hat{\mathbf{q}}(0), \hat{\mathbf{q}}(0) \rangle}{\langle \hat{\mathbf{q}}(0), \hat{\mathbf{q}}(0) \rangle} = \frac{\langle \Phi^*(t) \Phi(t) \hat{\mathbf{q}}(0), \hat{\mathbf{q}}(0) \rangle}{\langle \hat{\mathbf{q}}(0), \hat{\mathbf{q}}(0) \rangle}, \quad (5)$$

¹Inversion is permissible in compressible flow due to the nonsingular nature of $\mathbf{B}$. In the incompressible limit, desingularization techniques also exist, e.g., the penalty method associated with finite-element spatial discretization (Ding & Kawahara 1998) and the utilization of divergence-free bases in a spectral context (Karniadakis & Sherwin 2005).

which permits the study of both modal and nonmodal perturbation growth in a unified framework. Implicit here is the definition of an inner product, $\langle \cdot, \cdot \rangle$, and the associated adjoints Φ^* and $\mathbf{C}^*$ of the matrices Φ and $\mathbf{C}$, respectively (Morse & Feshbach 1953). We complete the discussion by introducing the singular value decomposition of the propagator operator

$$\Phi(t) \equiv e^{\mathbf{C}t} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*. \quad (6)$$

Here the unitary matrices $\mathbf{V}$ and $\mathbf{U}$ comprise (as their column vectors) initial and final states, respectively, as transformed by the action of the propagator operator, and $\mathbf{\Sigma}$ is diagonal and contains the growth σ associated with each initial state as the corresponding singular value. Much like the 1D basic flow case, the singular value decomposition may be utilized to compute optimal perturbations. We note also that the operator $\Phi^*\Phi$ appearing in Equation 5 is symmetric, which has important consequences for its computation, as discussed below.

Eriksson & Rizzi (1985) discussed for the first time in the context of global instability analysis the approximation and computation of the matrix exponential. The two classic articles by Moler & van Loan (1978, 2003), 25 years apart, are essential reading in this context. More recently, an approach for the computation of the propagator has been described by Schulze et al. (2009).

If the basic flow is unsteady, with an arbitrary time dependence, then the propagator operator $\Phi(t)$ may also be defined, and Equation 4 may be generalized as

$$\hat{\mathbf{q}}(t_0 + \tau) = \Phi(\tau)\hat{\mathbf{q}}(t_0). \quad (7)$$

Here the propagator may be understood as the operator evolving the small-amplitude perturbation from its state at time t_0 to a new state at time $t_0 + \tau$. If the time dependence of the basic state is periodic with period T , $\forall t_0 : \bar{\mathbf{q}}(\mathbf{x}; t_0 + T) = \bar{\mathbf{q}}(\mathbf{x}; t_0)$, the propagator is denoted as the monodromy operator and it is also T periodic, $\Phi(t_0 + T) = \Phi(t_0)$. It is defined by (Karniadakis & Sherwin 2005)

$$\Phi = \exp \left[\int_{t_0}^{t_0+T} \mathbf{C}(\bar{\mathbf{q}}(\mathbf{x}; t')) dt' \right]. \quad (8)$$

Solutions to the instability problem, indicating the development of small-amplitude perturbations during one period of evolution, are obtained through Floquet theory, which seeks the eigenvalues of the monodromy operator, also known as Floquet multipliers, μ . To this end, the monodromy operator is evaluated at time T , and the EVP

$$\Phi(T)\hat{\mathbf{q}} = \mu\hat{\mathbf{q}} \quad (9)$$

is solved. The Floquet multipliers can also be expressed in terms of the Floquet exponents, γ , as $\mu = e^{\gamma T}$, which identifies $|\mu| = 1$ as a bifurcation point and indicates that

$$|\mu| < 1 : \text{periodic flow stability}, \quad (10a)$$

$$|\mu| > 1 : \text{periodic flow instability}. \quad (10b)$$

1.3. Terminology

A generally accepted terminology exists for 1D parallel or axisymmetric flows, in which $\bar{\mathbf{q}} = \bar{\mathbf{q}}(y)$, or $\bar{\mathbf{q}} = \bar{\mathbf{q}}(r)$, with y and r denoting wall-normal and radial spatial coordinates, respectively. Here one refers to local modal or nonmodal analysis, depending on whether the EVP or the IVP is solved, respectively. The monograph by Schmid & Henningson (2001) provides a complete and up-to-date account of flow instability in this limit. In the interesting but particular case of boundary-layer instability, a separation may be considered between the scales on which the basic

boundary-layer flow and the small-amplitude perturbations develop. Analysis in this situation is sometimes referred to as nonlocal instability and is based on the parabolized stability equations (PSE) (Herbert 1997). However, a newcomer to instability analysis of multidimensional basic states may well be confused by the existence in the literature of a single term, global (in)stability, used to describe three different theoretical concepts.

Joseph (1966) first introduced the term global stability analysis in a methodology that monitors perturbation energy at all times and establishes lower bounds for flow stability. This approach has been widely used since the 1960s (e.g., Homsy 1973, Reddy & Voyé 1988), invariably treating 1D basic flows $\bar{\mathbf{q}}(y)$. No work is known to the author in which energy stability limits have been predicted in a complex flow, although perturbation energy has been monitored in a lid-driven cavity flow by Albensoeder et al. (2001).

Work in the early 1980s (Pierrehumbert & Widnall 1982, Eriksson & Rizzi 1985, Pierrehumbert 1986) marked the beginning of the second class of global instability approaches, those based on the solution of the EVP pertaining to essentially 2D basic states, $\bar{\mathbf{q}} = \bar{\mathbf{q}}(x, y)$. In the past decade, solutions of the IVP in such flows have appeared in the literature, whereas the 3D EVP associated with basic flows $\bar{\mathbf{q}} = \bar{\mathbf{q}}(x, y, z)$ has also been solved for the first time by Tezuka & Suzuki (2006). The current literature collectively refers to the analysis based on the solution of the EVP or the IVP, with $\bar{\mathbf{q}}$ a function of two or three spatial coordinates, as global instability theory, with global modes denoting eigenvectors $\hat{\mathbf{q}}$, solutions of Equation 3.

Finally, the roots of a third global instability theory lie in absolute/convective instability analysis ideas, introduced in the 1960s in plasma physics (Briggs 1964) and employed to analyze parallel and weakly nonparallel flow instability since the mid-1980s. Huerre & Monkewitz (1990) discussed this third approach for parallel and weakly nonparallel flows, following the definition of global modes by Chomaz et al. (1988). (Details are given in Chomaz 2005.) Experience amassed in the past decade suggests that the latter two global analysis concepts may lead to qualitatively, but not necessarily quantitatively, consistent results. Furthermore, the identification of solutions of Equation 2 or 3 pertinent to essentially nonparallel 2D or 3D flows with global modes in the sense of Chomaz et al. (1988) is not necessarily true.

To avoid potential confusion, the terms BiGlobal and TriGlobal instability analysis have been proposed (Theofilis et al. 2001) to describe (modal or nonmodal) analyses of 2D and 3D basic states, respectively. Although this terminology has been widely used in the literature (Karniadakis & Sherwin 2005, Longueteau & Brazier 2008, Piot et al. 2008, Groskopf et al. 2010), another alternative term, direct instability analysis, has also been put forward by Barkley et al. (2008), whereas Carpenter et al. (2010) introduced another term, the harmonic linearized Navier Stokes. The need for terminology clarifying the meaning of global instability analysis acknowledges the potential for confusion with the all-inclusive global terminology; the present review uses the BiGlobal and TriGlobal terminology.

Nonmodal BiGlobal instability analysis is an emerging field of research, and the bulk of global instability analyses performed in its three-decade history employ the numerical solution of the BiGlobal EVP. Section 2 introduces numerical tools for spatial discretization and eigenspectrum computation applicable to both modal and nonmodal instability analysis. Section 3 briefly presents recent results of modal and nonmodal BiGlobal, as well as modal TriGlobal, analyses. There is a fast-growing body of almost exclusively theoretical literature on instability (and control) of multidimensional basic states. A set of archival-quality peer-reviewed research papers has been collected in a special journal issue dedicated to the subject (Theofilis & Colonius 2010) and provides a good overview of current activities in the area of instability analysis and control of multidimensional flows.

BiGlobal: analysis of global instability in a 3D domain with two inhomogeneous and one homogeneous directions, with the latter treated as periodic; no multiple-scales assumption is invoked

TriGlobal: analysis of global instability in a 3D domain with three inhomogeneous spatial directions

2. NUMERICAL CONSIDERATIONS

In the broadest sense, interest in performing global instability analysis is in the identification (and possibly control) of physical mechanisms associated with amplification of small-amplitude perturbations superposed upon flows over or through complex geometries. From a numerical point of view, the prime relevant considerations are (a) whether one should analyze the short-time (IVP) or long-time (EVP) limit, leading to symmetric or unsymmetric matrices, respectively; (b) whether reductions of the systems of equations addressed are possible, e.g., by exploiting symmetries or by solving a system in which a smaller number of equations (and higher degree of differentiation) is involved; and (c) whether real or complex matrices are involved. In addition, decisions must be made regarding the spatial discretization methodology to be employed and whether linear algebra operations are to be performed in a dense or sparse matrix context.

These considerations are significant as, even with present-day supercomputing hardware capabilities, the size of matrices discretizing the linearized equations of motion, when the latter are formed, can be formidable. A rule of thumb presented by Theofilis (2003) is that local, BiGlobal, and TriGlobal instability can be described by matrices having their respective size, measured in MB, GB, and TB. In this context, the need for low-/reduced-order models to represent the full system is as true today for TriGlobal analysis as it was when it was first introduced for BiGlobal instability by Noack & Eckelmann (1994).

2.1. Spatial Discretization

When discretizing the spatial operator in global instability analysis, a fundamental choice is between high- and low-order methodologies. High-order methods are preferable when seeking to minimize the number of discretization points necessary for convergence. In solution approaches in which memory is not a predominant issue, either because sparse linear algebra techniques are used in conjunction with matrix formation and storage or because the matrix is not formed at all, low-order methods may also be employed. In this situation, resolution may, in principle, be increased to levels substantially higher than those needed by high-order methods until convergence is achieved.

2.1.1. Spectral methods. Spectral methods (Canuto et al. 2006) have had a prominent role in early global instability analyses and are presently as useful as ever. They were used in the first inviscid global analysis works, namely those by Pierrehumbert & Widnall (1982) and Pierrehumbert (1986), who solved the perturbed form of the Euler equations, and by Henningson (1987), who solved the 2D Rayleigh equation; in the first viscous instability analysis of an open flow, namely the wake of a circular cylinder (Zebib 1987), and that of a closed system, the grooved channel (Amon & Patera 1989); and in numerical work associated with the first theoretically founded control of global flow instability by Hill (1992). Spectral collocation is still used, e.g., in the massively parallel computations of the BiGlobal eigenvalue spectrum in a plane (Rodríguez & Theofilis 2009) and analytically transformed domains (Kitsios et al. 2009) and in the TriGlobal analysis of Bagheri et al. (2009c).

The combination of spectral methods with multidomain techniques (Demaret & Deville 1991, Deville et al. 2002) has been demonstrated for BiGlobal instability by De Vicente et al. (2006) and can serve to discretize geometries decomposable in regular subdomains. The potential of multidomain techniques has been exploited by Robinet (2007) for external aerodynamics and by Merzari et al. (2008) in an application arising in nuclear-reactor safety. The spectral multidomain methodology is a particular case of the spectral-element approach, which has been well exploited for global instability analysis. Spectral element techniques have been introduced to global

instability analyses of Newtonian (Amon & Patera 1989, Barkley & Henderson 1996) and non-Newtonian fluids (Fiétier 2003), in which regular geometries have been discretized by structured-grid approaches. Alternatively, the more flexible spectral/*hp*-element method (Karniadakis & Sherwin 2005), which permits space tessellation using hybrid structured and unstructured meshes comprising rectangular and/or triangular elements, has been employed to study modal global instability of flow around a NACA-0012 airfoil (Theofilis et al. 2002), as well as modal and nonmodal instability of flow through a geometry of physiological relevance (Sherwin & Blackburn 2005) and around a cascade of low-pressure turbine blades (Abdessemed et al. 2009, Sharma et al. 2010).

2.1.2. Finite-element and finite-volume methods. The realization that unstructured meshes are a convenient spatial discretization methodology for the complex geometries potentially encountered in a global instability analysis naturally led to the use of finite-element methods since the earliest days of such analyses. Jackson (1987) discretized the spatial operator describing instability in the wake of a circular cylinder by finite-element methods. The same spatial discretization has been employed for the solution of the EVP in the wake of spheres and discs (Natarajan & Acrivos 1993), in lid-driven cavity flows (Ding & Kawahara 1998), counterflowing jets (Pawlowski et al. 2006) and systems of trailing vortices (González et al. 2008), in modal and nonmodal BiGlobal analysis of S-shaped duct flows (Marquet et al. 2008), and in TriGlobal modal analysis of instability in the wake of cylinders and spheres (Morzyński & Thiele 2008).

Second-order finite-volume methods were introduced by Dijkstra (1992) to discretize the equations describing the global stability of cellular solutions in Rayleigh-Benard-Marangoni flows. Albensoeder et al. (2001) also used finite-volume spatial discretization in their stability analysis of a square lid-driven cavity and independently recovered results in excellent agreement with the reference spectral collocation solution of the same problem (Theofilis 2000).

2.1.3. Finite-difference methods. Standard second-order central finite differences have been utilized in circular cylinder wake analysis by Wolter et al. (1989) and Morzyński & Thiele (1991). Such methods are still in use, e.g., in instability and sensitivity analyses of incompressible flows over a forward-facing step (Marino & Luchini 2009) and in a cylinder wake (Giannetti & Luchini 2007), the latter analysis employing an immersed-boundary approach (Peskin 1977, Mittal & Iaccarino 2005). Thermocapillary flow instabilities, studied by spectral methods by Hoyas et al. (2004), have also been analyzed using standard second-order finite-difference methods on a staggered grid by Shiratori et al. (2007). Finally, Giannetti et al. (2009) performed the first TriGlobal instability analysis of incompressible flow in a cubic lid-driven cavity using second-order accurate, staggered finite-differencing of the 3D linearized Navier-Stokes operator.

In compressible flow, Eriksson & Rizzi (1985) employed variable-order finite-difference approximations in their inviscid analysis of transonic flow over a NACA-0012 airfoil. High-order compact finite-difference methods have been employed for the solution of compressible global instability problems by Bres & Colonius (2008) in an open cavity and by Mack & Schmid (2010) in a swept leading-edge boundary layer. Also in compressible flow, but in contrast to all previous global instability work, which analyzed steady or time-periodic laminar solutions to the Navier-Stokes equations at moderate Reynolds numbers, Crouch et al. (2007) presented the first global instability analysis of turbulent transonic flow at flight Reynolds numbers, in which the shock system is embedded as an integral part of the analysis. The presence and proper capturing of the entire shock system in the footprint of the global eigenmodes led to the use of a combination of (fourth-order) central and (third-order) upwind finite-differencing.

Low-order accuracy is typically compensated for by a high number of coupled degrees of freedom, presently of $O(10^6)$, to achieve convergence of global instability results. This number, as well

as those quoted for the spectral methods, is a snapshot of current computing technology and will clearly increase in the future. Nevertheless, this indicates that, since its early days, global instability analysis has been associated with the limits of large-scale scientific computation. Moreover, in the majority of cases, the success of the analysis strongly depends on the efficiency of the algorithm for the recovery of the eigenspectrum subsequently utilized. Such algorithms are reviewed in Section 2.2.

2.1.4. Some comments on boundary conditions. Here we briefly comment on boundary closures for the elliptic operators of global instability analysis. The only situation in which these conditions are clear is at solid walls where, depending on the type of analysis performed and the variable in question, viscous or inviscid conditions may be employed. At the far field of an open system, the situation is analogous with that of local instability analysis; vanishing of perturbations is one option, although analytic boundary conditions may also be derived and utilized in a receptivity context (Tumin & Fedorov 1984). Inflow and outflow boundaries of open systems receive distinct treatment in general. At inflow boundaries, homogeneous Dirichlet boundary conditions may be imposed, based on an inviscid, high-Reynolds number reasoning (Morzyński & Thiele 1991, Natarajan & Acrivos 1993) or to prevent incoming viscous disturbances (Theofilis et al. 2000). Several soft-type outflow boundary conditions have been attempted at the outflow boundary, such as the standard zero-normal-stress conditions typical of finite-element computations (Jackson 1987), the parabolized outflow conditions (Tomboulides et al. 1993), and linear extrapolation (Theofilis et al. 2000, 2003). They typically lead to the formation of a narrow unphysical region adjacent to the outflow boundary, which is discarded when postprocessing the results while the main resolved features of the field remain unchanged. Although it may only be justified by a posteriori inspection of the independence of both the eigenspectrum and the spatial distribution of the amplitude functions on parameters such as the extent of the domain, this heuristic approach has led to consistent global instability analysis results in a variety of bluff-body flows, as well as several discoveries in boundary-layer-type flows, such as that of the stationary 3D global mode of laminar separation (Theofilis et al. 2000) and the polynomial structure of global eigenmodes in an incompressible swept leading-edge flow (Theofilis 1997, Theofilis et al. 2003).

2.2. Eigenspectrum Computation

Intense activity spanning the second half of the twentieth century resulted in the development of both direct and iterative algorithms for the solution of EVPs arising in computational mechanics (Cliffe et al. 1993, Watkins 1993, Golub & van Loan 1996). However, computing hardware has experienced profound changes in the past three decades, and, in the author's view, the application of algorithms developed for use on serial machines with a limited amount of shared memory may be an artificial barrier that needs to be broken for the analysis to reach its full potential. A related point is the warning issued by Morzyński & Thiele (2008) regarding the applicability to a given hardware of published results on the performance of libraries obtained on different (typically older) machines.

After briefly describing full-spectrum computations, this section mainly covers iterative methods for global instability analysis. The discussion is a conscious attempt to expose the simplicity of the tools and hopefully reverse the observed trend in recent global instability literature, whereby an ever-increasing section of the community relies on library software incorporating eigenspectrum computation algorithms. In the process, the flexibility offered by own-developed software is lost, the reliability of the results entirely depends on the quality of the implementation, and the learning curve is dominated by a package that may be inefficient or obsolete on next-generation hardware.

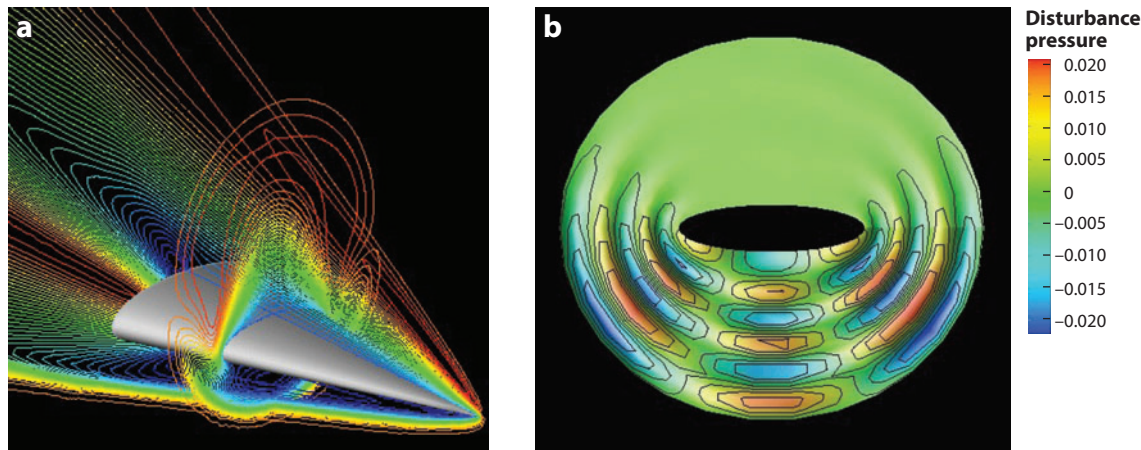


Figure 1

(a) Streamwise velocity component of the basic flow. (b) Pressure perturbation of traveling neutral eigenmode in inviscid global instability analysis of Mach 4 flow over an elliptic cone at an angle of attack. Figure taken from Theofilis (2002).

2.2.1. Full eigenspectrum computation. Full eigenspectrum computation based on the QZ algorithm (Golub & van Loan 1996) has been employed for the solution of multidimensional EVPs since their first appearance in the literature. Early 2D EVP solutions (Pierrehumbert & Widnall 1982, Pierrehumbert 1986, Zebib 1987, Wolter et al. 1989, Tatsumi & Yoshimura 1990) used full eigenspectrum computation.² When feasible, full eigenspectrum computation is still used (Merzari et al. 2008, Swaminathan et al. 2010), as it provides straightforward access to otherwise tedious-to-obtain insight into the eigenspectrum of a new problem.

Although this computation is by far the easiest to implement, only requiring the set up of the discretized version of the matrices describing the generalized EVP, the dense matrix operations utilized make it prohibitively expensive when leading dimensions, $n \sim O(10^5 - 10^6)$, are encountered in coupled discretization of multidimensional EVPs: Computing time scales as $O(n^3)$ and four $O(n^2)$ matrices need to be stored, two associated with the generalized EVP [even though one of them is (block) diagonal in the incompressible case] and two auxiliary matrices. The intrinsically serial nature of full-spectrum computation algorithms hardly helps alleviate any of these drawbacks by parallelization. The use of distributed-memory machines is practically precluded by the need for storage of and continuous operation on the matrices.

However, three decades after the first global instability analysis appeared in the literature, the vast majority of hydrodynamic and aeroacoustic instabilities in complex geometries remain unexplored. The virtue of full-spectrum computations is that they provide the safest means of identifying all eigenmodes. Situations have arisen in the literature in which iterative eigenspectrum computations led to the most important branch of eigenvalues being missed (Ding & Kawahara 1998). Of no less significance is the fact that full-spectrum information aids in the classification of different branches of eigenvalues, corresponding to different physical phenomena (e.g., the instability of hydrodynamic or aeroacoustic origin). For example, the results of **Figure 1** show a member of one acoustic branch identified in the inviscid global eigenspectrum of Mach 4 flow

²Full eigenspectrum computation is also referred to as global eigenvalue problem solution (Malik & Orszag 1987), although work in that vein deals with local instability analysis.

over an elliptic cone (Theofilis 2002). In this context, the argument often heard, echoing its first appearing in Edwards et al. (1994), regarding the importance of the leading eigenvalues alone is weakened. Full-spectrum computations are recommended whenever available computing hardware permits them and their computing requirements are reasonable. Finally, we note that in the context of full-spectrum computation it is straightforward to assess the effect of grid resolution. Although the discretized approximation of the continuous spectrum will always be under-resolved, the most interesting discrete members of the eigenspectrum are the first to converge, provided sufficient grid points are available to resolve their structure. Further grid refinement only serves to confirm the accuracy of the leading discrete modes and recover additional discrete family members.

2.2.2. Iterative global instability analyses. Notwithstanding the above discussion, full-spectrum computation is only possible in a few situations. Iterative eigenspectrum recovery drastically reduces hardware requirements and is by far the most widely employed tool of global instability analysis, usually in conjunction with some form of spectral transformation. Below we define the rather wide concept of iteration somewhat more precisely by distinguishing between the broad field of iterative approaches for global instability analysis based on time-stepping and the narrower area of iterative methods for the solution of the EVP. The latter area, within which most global instability analysis work has been performed, is discussed first. This dichotomy should not be understood as representing distinct classes of iterative approaches, as elements found in both classes have often been mixed together to build a single iterative algorithm.

Spectral transformations. When possible, the first step in preprocessing the eigenspectrum sought is to remove spurious eigenmodes by constructing a set of basis functions that satisfy the boundary conditions (Zebib 1987, Tatsumi & Yoshimura 1990, Uhlmann & Nagata 2006; see Boyd 1989 for details in a spectral discretization context).

Furthermore, spectral transformations may be introduced to analytically modify the eigenvalue spectrum and facilitate the extraction of the desired eigenvalues by confining the infinite branches of eigenvalues into a localized region in space. The latter branches can be avoided by some shift strategy, such that a subsequent iteration may target the desired discrete eigenmodes. This procedure, which is intended to improve convergence of the iteration by the appropriate choice of the parameters involved, is sometimes abbreviated as preconditioning. **Table 1** presents spectral transformations commonly used in global analysis, which convert the original generalized EVP given in Equation 3 into the standard problem

$$\hat{\mathbf{A}}\mathbf{x} = \lambda\mathbf{x}. \quad (11)$$

Shown are an $O(k)$ polynomial approximation to the matrix exponential (Eriksson & Rizzi 1985), the well-known in the context of the Arnoldi (1951) algorithm shift-and-invert transformation with shift parameter c_1 , the less utilized bilinear transformation (Christodoulou & Scriven 1988), and two- (Dijkstra et al. 1995) and three-parameter (Morzyński et al. 1999) variants of the Cayley transformation, with c_2 and c_3 related in this case to the zero and the pole of the transformation. The transformed eigenvalues are also presented in **Table 1** in a form that clearly shows the confinement of the spurious/infinite eigenvalues of the original problem. For illustrative purposes, **Supplemental Appendix 1** discusses the effect of the Arnoldi and a two-parameter Cayley transform on the eigenspectra of plane Poiseuille and rectangular duct flows (follow the Supplemental Material link from the Annual Reviews home page at <http://www.annualreviews.org>). In-depth discussions of general spectral transformations are provided by Christodoulou & Scriven (1988), Cliffe et al. (1993), and Meerbergen et al. (1994).

Table 1 Spectral transformations commonly utilized in global instability analysis

Name	Transformed matrix	Transformed eigenvalue	First application to global analysis
Polynomial approximation to exponential transformation	$\hat{\mathbf{A}} = [\mathbf{B} - (1 - c_2)c_3(\mathbf{A} - c_1\mathbf{B})]^{-k}[\mathbf{B} + c_2c_3(\mathbf{A} - c_1\mathbf{B})]^k$	$\lambda = \left[\frac{\frac{1}{\omega} + (1 - \frac{c_1}{\omega})c_2c_3}{\frac{1}{\omega} - (1 - \frac{c_1}{\omega})(1 - c_2)c_3} \right]^k$	Eriksson & Rizzi (1985)
Shift and invert	$\hat{\mathbf{A}} = (\mathbf{A} - c_1\mathbf{B})^{-1}\mathbf{B}$	$\lambda = \frac{1}{\frac{1}{\omega} - \frac{c_1}{\omega}}$	Christodoulou & Scriven (1988)
Bilinear	$\hat{\mathbf{A}} = (\mathbf{B} - c_1\mathbf{A})^{-1}(\mathbf{B} + c_1\mathbf{A})$	$\lambda = \frac{\frac{1}{\omega} - 1}{\frac{1}{\omega} + 1}$	Christodoulou & Scriven (1988)
Two-parameter Cayley	$\hat{\mathbf{A}} = -[\mathbf{A} - (c_1 + c_2)\mathbf{B}]^{-1}[\mathbf{A} + (c_1 - c_2)\mathbf{B}]$	$\lambda = \frac{\frac{c_1 - c_2}{\omega} + 1}{\frac{c_1 + c_2}{\omega} - 1}$	Cliffe et al. (1993)
Three-parameter Cayley	$\hat{\mathbf{A}} = (\mathbf{A} - c_2\mathbf{B})^{-1}[(1 - c_3)\mathbf{A} - (c_1 - c_2c_3)\mathbf{B}]$	$\lambda = \frac{1 - c_3 + \frac{c_2c_3 - c_1}{\omega}}{1 - \frac{c_2}{\omega}}$	Morzyński et al. (1999)

Iterations for the recovery of the eigenspectrum. Inverse iteration, which focuses on computation of the leading eigenmode (Golub & van Loan 1996), was introduced into global instability analysis by Jackson (1987) in his study of instability in the wake of a circular cylinder. This approach was also used for the same problem by Morzyński & Thiele (1991) as an alternative to the full-spectrum computation of Zebib (1987) and Wolter et al. (1989). Presently variants of inverse iteration have been devised by Marino & Luchini (2009) and Giannetti & Luchini (2007) in their analyses of instability and sensitivity in forward-facing step flow and cylinder wake, respectively, as well as by Wanschura et al. (1995) and Shiratori et al. (2007) in their studies of thermocapillary instabilities.

Having utilized a Galerkin method and an expensive splines-based approach for the recovery of eigenvalues in their earlier work, Dijkstra (1992) and Dijkstra et al. (1995) have employed the two-parameter Cayley transform of **Table 1** and the simultaneous iteration technique, originally proposed by Stewart & Jennings (1981) for large sparse real matrices, to obtain leading eigenvalues by power iteration. A common idea in these works, expanded upon in the original references, is filtering out eigenvalues of small magnitude. Morzyński et al. (1999) have also used the simultaneous iteration technique in conjunction with the preconditioned three-parameter Cayley transform of **Table 1** in their analyses of incompressible cylinder wake instability, whereas Mack & Schmid (2010) demonstrated another variant of the Cayley transform in the global analysis of compressible swept leading-edge flow.

For the computation of the global eigenspectrum itself, the most used iterative approach has been the Arnoldi (1951) algorithm, one of the best-known members of the Krylov subspace iteration methods, described in detail by Saad (1980). Natarajan (1992) first discussed the Arnoldi algorithm for local linear stability analysis, prior to successfully applying it to the prediction of instability in the wake of a sphere (Natarajan & Acrivos 1993). Other early analyses reporting the use of the Arnoldi algorithm included the study of the instability of flow over riblets (Ehrenstein 1996) and that in an incompressible swept leading-edge boundary-layer flow (Theofilis 1997).

The list of applications that have used the Arnoldi algorithm in global stability analysis is too long to be cited here, the algorithm being simple enough to be written symbolically and coded in a small number of lines. Much like all algorithms of the Krylov class, the main element of the Arnoldi iterative process is the generation of a (Krylov) subspace, $\mathcal{K}_m$, of dimension m , by repeated

application of a (time-independent) discretized matrix (the Jacobian) $\mathbf{M}$, to an arbitrary (usually, but not necessarily, unit-length) initial vector $\mathbf{q}_0$,

$$\mathcal{K}_m = \text{span}\{\mathbf{q}_0, \mathbf{M}\mathbf{q}_0, \mathbf{M}(\mathbf{M}\mathbf{q}_0), \dots, \mathbf{M}^{m-1}\mathbf{q}_0\}. \quad (12)$$

The key idea is that the set of vectors generated by the iteration spans a relatively small Krylov subspace; in that small subspace a good approximate solution of the original large EVP can be found by standard full-spectrum methods.

The Arnoldi iteration may be embedded in a unified framework for both modal and nonmodal analysis. For modal analysis, $\mathbf{M}$ may be taken as either of $\hat{\mathbf{A}}$ or Φ , defined in Equations 11 or 4, respectively, as their eigenvalue systems Λ and Σ are directly related by $\Sigma = \exp(\Lambda t)$ (Eriksson & Rizzi 1985). Alternatively, taking $\mathbf{M}$ equal to the normal matrix $\Phi^*\Phi$ defined in Equation 5, nonmodal analysis may be performed.

In the latter case, the Arnoldi algorithm may not be as efficient in accomplishing this task as another subspace iteration algorithm, namely that of Lanczos (1950). Its departure point is again the construction of a Krylov subspace using Equation 12, although the technical details differ from those of the Arnoldi algorithm, because efficient tridiagonal matrix operations are performed in the Lanczos algorithm, as opposed to the wide-banded or dense matrix operations required by the Arnoldi algorithm. Versions of the Lanczos algorithm also exist for the general non-Hermitian case. Nayar & Ortega (1993) discussed one in the context of local modal analysis of non-Hermitian matrices, such that direct comparisons with the results of the Arnoldi algorithm were possible, and provided an elaborate discussion on filtering, which permits the distinction between true and spurious eigenvalues.

Whereas early global instability practitioners implemented the Arnoldi iteration, many works appearing since the end of the past century have utilized the well-tested ARPACK library to perform the iteration. Lehoucq & Salingier (2001) and Pawlowski et al. (2006) have reported the successful use of the parallelized version of the same library. Technical details of parallelization of the Arnoldi algorithm itself have been discussed by Rodríguez & Theofilis (2009), who analyzed each element composing the entire eigenspectrum recovery procedure as a fraction of the total cost of the computation and discussed scalability of the parallelized Arnoldi algorithm on distributed-memory supercomputers. Wall-clock timing scaled by the (appropriate for dense algebra operations) cube of the leading dimension of the matrix is presented as a function of the number of processors utilized for the solution of Equation 11 on the JUGENE machine in **Figure 2**.

Alternatively, numerical methods for generalized eigenspectrum computation of large symmetric matrices have reached a rather advanced state of development. Details on the basic shift-invert block Lanczos algorithm may be found in Grimes et al. (1994), whereas recent developments applying to matrices with leading dimensions of $O(10^5-10^6)$ have been discussed by Arbenz et al. (2005).

Finally, a (non-Krylov) subspace iteration method is the Jacobi-Davidson QZ algorithm (Sleijpen & van der Vorst 2000). It treats both the linear and polynomial EVPs in a unified framework, which can be useful when dealing with spatial BiGlobal stability. The Jacobi-Davidson QZ algorithm has been analyzed and implemented for a Rayleigh-Benard convection problem by Van Dorsselaer (1997) and for the incompressible swept Hiemenz flow by Heeg & Geurts (1998). Borges & Oliveira (1998) have discussed the properties of a parallel implementation of a Davidson-type algorithm and compared its performance with those delivered by the parallel implementation of the ARPACK library discussed above. Hwang et al. (2010) described a recent parallel implementation of the Jacobi-Davidson algorithm, which permits efficient recovery of eigenvalues of the Schrödinger equation using $O(10^7)$ coupled degrees of freedom, on a modest,

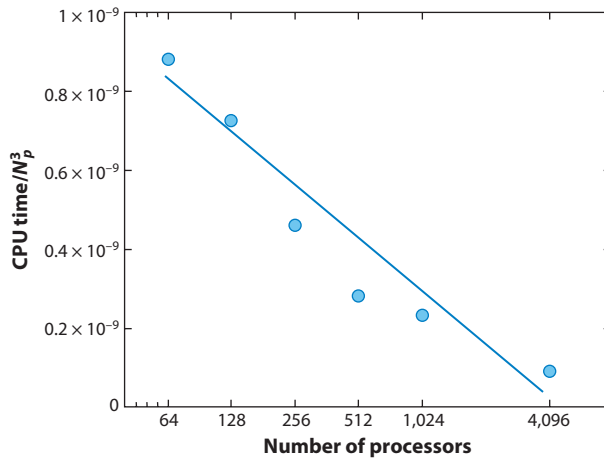


Figure 2

Wall-clock time scaled by the size of the (dense) matrix as a function of the number of processors utilized for the massively parallel numerical solution of the BiGlobal eigenvalue problem (Rodríguez & Theofilis 2009) on the IBM Blue Gene/P computer at <http://www.fz-juelich.de>.

$O(10^2)$, number of processors at the cost of $O(10)$ min of wall-clock time; such resolutions and performance have not been demonstrated for global instability analysis.

We note that the spectral transformations shown in **Table 1** all require computation of the inverse of a matrix. This turns out to be one of the more costly elements of the analysis, due to the large size of the matrices involved. In fact, instead of full inversion, the standard approach is to perform an LU decomposition, save the factors, and solve the linear systems appearing within the iterative eigenspectrum computation by forward and backward solves, updating the right-hand-side terms during the iteration. An alternative and more efficient approach has been proposed by Giannetti et al. (2009), based on approximate matrix inversion. Yet another point of view may be taken on iteration, which altogether circumvents matrix formation. Approaches following this path in the literature are denoted time-stepping, matrix-free, Jacobian-free, and snapshot methods, although each of these terms may carry additional connotations.

2.2.3. On time-stepping iterative approaches. The departure point of time-stepping approaches is casting the equations of motion as an IVP,

$$\frac{\partial \mathbf{q}}{\partial t} = \mathbf{F}(\mathbf{q}). \quad (13)$$

Although $\mathbf{F}$ could be taken to describe the linearized operator as in Equation 2, as is done below, in the most general case it describes the spatial operator pertinent to the complete field, $\mathbf{q} = (\rho, u, v, w, p)^T$, as for direct numerical simulation, large-eddy simulation, or Reynolds-averaged Navier-Stokes computation. Next $\mathbf{F}$ is perturbed in the direction of an arbitrary vector $\mathbf{q}_0$; the existence of the limit

$$\lim_{|\epsilon| \rightarrow 0} \frac{\|\mathbf{F}(\mathbf{q} + \epsilon \mathbf{q}_0) - [\mathbf{F}(\mathbf{q}) + \epsilon \mathbf{L}(\mathbf{q})\mathbf{q}_0]\|}{|\epsilon|} = 0, \quad (14)$$

with $\epsilon \in \mathbb{R}$, defines the linear operator $\mathbf{L}(\mathbf{q})$, known as the Fréchet derivative (Keller 1975). When $\mathbf{F}(\mathbf{q})$ describes a system of algebraic equations, as the case is when Equation 13 is discretized

and boundary conditions are incorporated (Boyd 1989), the Fréchet derivative is the Jacobian, $\mathbf{L}(\mathbf{q}) = \partial \mathbf{F} / \partial \mathbf{q}$.

One could proceed by constructing (and storing) the Jacobian, in which case the formalism discussed above would follow. The key idea of Jacobian-free methods, originally suggested for global instability analysis by Eriksson & Rizzi (1985), is to utilize Equation 14 to solve for the Fréchet differential, $\mathbf{L}(\mathbf{q})\mathbf{q}_0$, and obtain (variable-order) approximations of the Jacobian by evaluations of the right-hand side of Equation 13 at the neighboring points $\mathbf{F}(\mathbf{q})$ and $\mathbf{F}(\mathbf{q} + \epsilon \mathbf{q}_0)$. Such evaluations are inherent to the tools already available in the algorithm (i.e., the code) utilized for the numerical solution of Equation 13. Eriksson & Rizzi (1985) also realized the need for a spectral transformation of the Jacobian matrix and proposed the polynomial approximation for the matrix exponential shown in **Table 1**, the effect of which is to promote the appearance of the interesting, most unstable/least stable eigenmodes, as opposed to those (possibly spurious ones) having the largest magnitude. Details on Eriksson & Rizzi's (1985) ideas on Jacobian approximation and exponential transformation were provided one decade later in the well-known works of Edwards et al. (1994) and Mamun & Tuckerman (1995).

In the particular case in which $\mathbf{F}$ represents the linearized equations of motion, a generalized Taylor expansion may be used to relate Equations 2 and 13. In time-stepping too, eigenspectrum computation may follow the construction of a Krylov subspace. However, in contrast with Equation 12, the subspace here is formed not by repeated application of the (same, time-independent) operator $\mathbf{M}$, but by evaluation of the (time-dependent) operator at successive equidistant instants of time,

$$\mathcal{K}_m = \text{span}\{\mathbf{q}_0, \mathbf{M}(\Delta t)\mathbf{q}_0, \mathbf{M}(2\Delta t)\mathbf{q}_0, \dots, \mathbf{M}((m-1)\Delta t)\mathbf{q}_0\}, \quad (15)$$

as discussed by Bagheri et al. (2009a). A Krylov subspace based on Equation 15 is a direct consequence of the definition of the Fréchet differential, $\mathbf{L}(\mathbf{q})\mathbf{q}_0$, which simply amounts to applying the code to advance the initial vector $\mathbf{q}_0$ in time and is a generalization of the definition based on Equation 12. As such, an approach based on Equation 15 is applicable to both steady and time-periodic basic states. In a manner analogous with that in the autonomous system case, modal or nonmodal analysis may be performed by taking $\mathbf{M}(t)$ equal to $\Phi(t)$ or $\mathbf{C}(t)$ for modal analysis and $\mathbf{M}(t) = \Phi^*(t)\Phi(t)$ for transient-growth studies.

It is rather surprising that the contributions of Eriksson & Rizzi (1985) have gone unnoticed for a quarter-century to all but two research groups in Japan. The work of Chiba (1998), which may be accessed in **Supplemental Appendix 2**, is little known, no doubt because it is written in Japanese. Chiba followed a Jacobian-free approach incorporating the exponential transformation and the Arnoldi algorithm to identify Hopf bifurcation in (steady) 2D square lid-driven cavity flow and performed Floquet analysis in the (time-periodic) wake of the cylinder, recovering results in agreement with the well-known work of Barkley & Henderson (1996). More recently, Tezuka & Suzuki (2006) expanded the ideas of Eriksson & Rizzi (1985) and the numerical tools of Chiba (1998) to perform the first-ever TriGlobal instability analysis, as discussed in Section 3.

Goldhirsch et al. (1987) presented another early time-stepping approach, although for local instability analysis, and recovered large parts of the eigensystem of the Orr-Sommerfeld equation by operations involving time-marching the IVP given in Equation 2, repeated applications of the linearized operator to an arbitrary initial vector, and simple computations for the eigenvalues and eigenvectors. Despite its elegance and simplicity, this approach has not been implemented for global instability analysis.

A unified discussion of time-stepping methods for both modal and nonmodal instability was recently presented by Barkley et al. (2008). Almost parenthetically, in view of the wide scope of the

issue of flow control, we mention here that matrix-free methods are also an enabling technology in (global) flow control. The interested reader is referred to the recent modal and nonmodal analysis of boundary-layer flow on a flat plate by Bagheri et al. (2009a,b), followed by control of instabilities in a unified matrix-free context.

Further details on Jacobian-free methods in a broad computational mechanics context may be found in the excellent review by Knoll & Keyes (2004), who discuss analyses of tokamak edge plasma, magnetohydrodynamics, and geophysical flows. A recent global instability analysis algorithm that incorporates a Jacobian-free approach, the implicitly restarted Arnoldi algorithm, a Cayley spectral transformation, and an essential for acceptable convergence preconditioner is the preconditioned Cayley-transformed implicitly restarted Arnoldi algorithm of Mack & Schmid (2010), which has been successfully applied to BiGlobal instability of a compressible swept leading-edge flow.

We mention here that three (true, in the sense that no symmetries of the flow field are exploited) of the four TriGlobal instability analyses presently available in the literature have been performed in a time-stepping framework, namely that of flow around a sphere and a prolate spheroid by Tezuka & Suzuki (2006), the jet-in-cross-flow analyses by Bagheri et al. (2009c), and the analysis of flow in a cubic lid-driven cavity by Giannetti et al. (2009). In contrast, the fourth TriGlobal analysis (second in order of appearance in the literature) is namely the work of Morzyński & Thiele (2008), who analyzed the instability of flow in the wake of 3D cylinders and spheres. This work is based on forming and (approximately) inverting the Jacobian matrix. The very existence of successful analysis methodologies based on both matrix-free and matrix-forming algorithms underlines the possible different paths forward and prevents drawing conclusions or making suggestions regarding the supposed superiority of a given approach.

Finally, the discussion above may naturally raise curiosity regarding the combination of algorithms and hardware technologies that enable global instability analysis. An extended discussion of this point is provided in **Supplemental Appendix 3**.

 **Supplemental Material**

3. RECENT HIGHLIGHTS OF GLOBAL INSTABILITY ANALYSIS

We now turn our attention to the presentation and critical discussion of recent significant discoveries of instability physics in generic complex geometries. We emphasize results delivered by the application of modal and/or nonmodal global instability analysis that appeared after Theofilis (2003). Apologies are offered to the authors whose interesting work could not be reviewed here owing to space limitations.

However, before entering a discussion of physical knowledge amassed by application of global linear theory in concrete flow applications, some qualitative discussion of the eigenfunction results of global modal analysis is presented. This was deemed necessary as the community has been accustomed to viewing linear perturbations as being composed of one-dimensional amplitude functions, extended harmonically in the other two spatial directions to reconstruct known three-dimensional small-amplitude disturbances, e.g., a Tollmien-Schlichting wave or a Kelvin-Helmholtz roll. In contrast with the local analysis, eigenfunctions resulting from a global linear instability theory depend in an inhomogeneous manner on two or three spatial directions, reflecting the inhomogeneity of the respective underlying basic states and making direct comparisons with three-dimensional eigenfunctions of local analysis impossible. However, such a comparison is possible, at least qualitatively, between results of local and global theories in the limit that spatially confined parts of the basic states addressed by BiGlobal or TriGlobal theory are (quasi-) parallel flows, such that (quasi-) local theory is also applicable. Four flow configurations in closed and open systems are utilized to elucidate these points in **Supplemental Appendix 4**.

The reader will notice a prominent absence, namely a reference to the current vigorous efforts to reconcile results of local and global theories on the well-studied flat-plate boundary layer. There are two good reasons for this omission. First, no work is known to the author that has dealt with the issue of the inflow and outflow boundary conditions in open systems in a manner that permits direct comparisons of local and global analysis results. Global analyses that mimic the assumptions of local theory, and duly recover results in excellent agreement with those of the Orr-Sommerfeld equation, are viewed as wasteful demonstrations of known physics, and may only be performed in order to validate the global instability analysis solvers. An example of this is the Blasius flow analysis of Rodríguez & Theofilis (2008) in which the boundary layer was kept artificially parallel and streamwise periodicity was imposed. Second, even if (or when) the problem of boundary conditions is solved, the tools for local modal and nonmodal linear analysis are, in the author's view, the appropriate means to pursue study of flow instability in the flat plate. This preference is based on the grounds of the orders-of-magnitude less computing effort required for local parallel and weakly nonparallel flow analysis of wave-like disturbances compared with the computational demands of global linear theory. As a final (and somewhat provocative) comment, claims in recent global instability literature on the agreement between amplitude functions delivered by the Orr-Sommerfeld equation and PSE analysis, on the one hand, and the wall-normal dependence of BiGlobal eigenfunctions at fixed streamwise locations on a flat plate, on the other hand, should be viewed with a fair amount of healthy skepticism. The spatial distribution of different amplitude functions in BiGlobal analyses of flat-plate boundary-layer flows obtained under a variety of inflow-outflow boundary conditions is qualitatively identical. The true power of global instability analysis is in situations in which the assumptions of local or weakly nonparallel theories are not applicable.

One such example is the TriGlobal modal instability analysis of Bagheri et al. (2009c), who used a time-stepping approach previously validated in several BiGlobal problems to analyze the 3D global instability of a jet in cross-flow. An interesting result in this work is the recovery and classification as 3D global eigenmodes of instabilities that have been known for a century by the application of local theory to simplified model basic flows—parts of the 3D basic flow analyzed—as well as the intricate modifications that such instabilities experience when embedded in a realistic 3D basic state. We now turn our attention to other flows in which the quasi-local theory is either conditionally or not at all applicable.

3.1. Duct Flows

For a while, Tatsumi & Yoshimura's (1990) pioneering global modal instability analysis of steady laminar flow in a rectangular duct and Gavrilakis's (1992) influential direct numerical simulation work on turbulence in the square limit of this geometry appeared to be distinct areas of research. In fact, the former work provided the first known global analysis results that were inconsistent with observation and experiment, predicting linear stability of square-duct flow at all Reynolds numbers. An analogous prediction followed in the related geometry of a duct of elliptical cross section (Kerswell & Davey 1996).

These are certainly not the first modal predictions that do not agree with experiment. One of the best known failures of local modal theory is the related Hagen-Poiseuille flow (Drazin & Reid 1981). Shortly after Tatsumi & Yoshimura's (1990) work appeared in the literature, local nonmodal analysis delivered its first predictions in parallel shear flows (Reddy et al. 1993), pointing to the role that transient growth may play in explaining several of the shortcomings of local modal linear theory. Of particular relevance to duct flow is its aforementioned axisymmetric Hagen-Poiseuille analog, nonmodal local instability analysis of which was performed by Reshotko & Tumin (2001).

It was thus a matter of extending the tools of nonmodal theory to the duct geometry and inquire into BiGlobal stability of duct flows from a linear, nonmodal point of view. This was accomplished by Galletti & Bottaro (2004), who formulated and solved a parabolized system of global stability equations (an approach now known as PSE-3D) for the recovery of optimal perturbations (Farrell & Ioannou 1996), incorporating turbulence in a triple-decomposition context, as proposed by Reau & Tumin (2002). The key results have been the demonstration of large, $O(Re^2)$, transient growth in the duct geometry and the identification of and proposed connection between secondary corner structures found in their analysis and in the turbulent simulation of Gavrilakis (1992). In a follow-up work, Biau et al. (2008) further elaborated on the issue of optimal perturbations in square-duct flow and identified structures reminiscent of those encountered in direct numerical simulations of turbulent flow (Uhlmann et al. 2007).

In summary, the square-duct geometry reminds us that, much like the better studied 1D basic/mean flows, in the context of global instability applicable to complex geometries as well, nonmodal (global) analyses should complement solutions of the (global) EVP, certainly in case the results of the latter approach disagree with the physical reality delivered by experiment or direct numerical simulation.

3.2. Lid-Driven and Open Cavity Flows

Whereas the large-aspect-ratio duct flow may be related to the classic channel boundary layer, the lack of homogeneous spatial directions in 2D lid-driven cavity flow makes the use of global linear analysis essential. By contrast, in open cavity flows instability in the shear layer emanating from the upstream corner has been analyzed with some success by local theory (Rowley et al. 2002). The two classes of cavity flows are also different in the boundary conditions applicable to the respective global instability analyses. Straightforward viscous boundary conditions may be applied to the wall-bounded lid-driven cavity, as opposed to the nontrivial boundary conditions applicable to the inflow and outflow boundaries of the open cavity. Recent progress in the analysis of both classes of cavity flows is summarized next.

3.2.1. Lid-driven cavities. In contrast with its failure in ducts, another class of closed flows, lid-driven cavities, is a prime example of successful predictions of modal global analysis. Lid-driven cavity flows were last reviewed by Shankar & Deshpande (2000), who provided a mostly phenomenological description of flow phenomena in two and three spatial dimensions. Practically all global instability analysis work has been performed after that review appeared, and the role of global eigenmodes in describing unsteadiness and three-dimensionality in this class of flows is now well-understood.

To the author's knowledge, Gresho et al. (1984) were the first to suggest a physical origin of instabilities observed in numerical simulations. They described their attempts to obtain a steady-state solution in the square cavity at $Re = O(10^4)$ as follows:

On occasion, however, the effects are quite noticeable and seem to be related to a sort of 'periodic' excitation of the system's 'normal modes'. This effect, which can probably also be regarded as a continuous application of a linear stability 'analysis' via small perturbations, is interesting but often annoying—especially when the flow is trying to approach a steady state. (Gresho et al. 1984)

Several authors have searched for Hopf bifurcation in a 2D square lid-driven cavity flow, solving the 2D BiGlobal EVP (one in which a spanwise wave-number parameter, $\beta = 2\pi/L_z = 0$, is considered, where L_z is the spanwise periodicity length). The most quoted 2D critical Reynolds

number is $Re_{2D} \approx 8,000$, with results being constantly refined. However, this discussion is largely academic, because 3D ($\beta \neq 0$) analysis has found the flow to be unstable at an order-of-magnitude lower Reynolds number.

The first accurate (but incomplete) 3D BiGlobal instability analysis was reported in a series of papers by Ding & Kawahara (1998). The complete global eigenmode information was first unraveled by Theofilis (2000), who identified the critical conditions at $(Re_{3D}, \beta) \approx (782, 15.4)$ and classified the 3D eigenmodes in terms of increasing critical Reynolds number: The leading eigenmode, S1, is a stationary instability, followed by three traveling modes, T1, T2, and T3, of which T2 corresponds to that discovered by Ding & Kawahara (1998). Independently, these results were arrived at and fully confirmed by Albensoeder et al. (2001) and many authors since. Parenthetically, we note that, from a numerical point of view, these results were confirmed by D. Barkley (private communication, 2000), who used the matrix-free time-stepper approach discussed above on a (past-decade-technology) laptop computer to obtain the results that Theofilis (2000) unraveled using a matrix-forming approach via full-spectrum and Arnoldi computations on a then top-of-the-range vector supercomputer.

More recently, the analysis of instability in lid-driven cavity flows of more complex cross-sectional shapes has been accomplished. 2D steady lid-driven cavity flow in an equilateral triangular domain has been analyzed with respect to its 3D instability by González et al. (2007), who established critical conditions for a particular regularization of the basic state. De Vicente (2010) found that the 2D L-shaped lid-driven cavity flow (Oosterlee et al. 1993) loses its stability against 3D BiGlobal eigenmodes at $Re \approx 650$, $\beta \approx 9.7$. The vorticity of the leading eigenmode is shown in Figure 3.

The first TriGlobal modal instability analysis of a cubic lid-driven cavity flow was accomplished recently by Giannetti et al. (2009), who devised a time-stepping algorithm, at the heart of which is an approximate matrix inversion, to unravel the leading eigenmodes of the 3D linearized

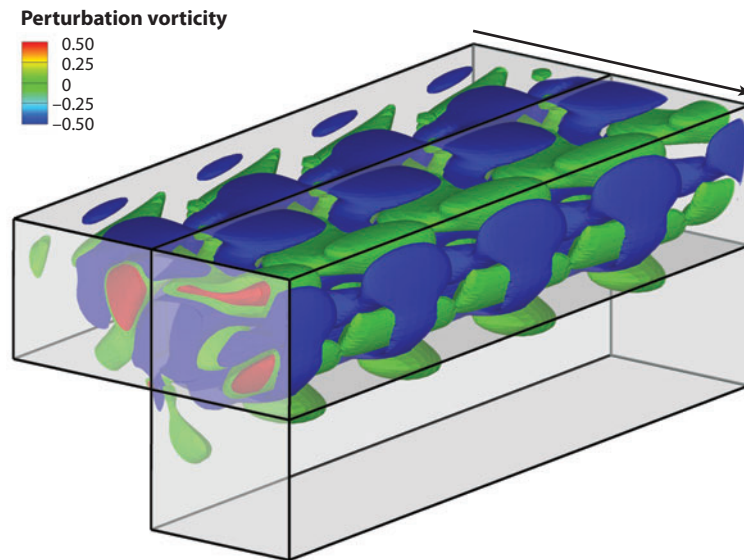


Figure 3

Isosurfaces of perturbation vorticity of the leading eigenmode in a lid-driven L-shaped cavity flow at $Re = 650$, $\beta = 9.7$. Figure taken from De Vicente (2010).

Navier-Stokes operator. Global instability analysis of the Stokes limit of the same flow was performed earlier by Leriche & Labrosse (2007).

3.2.2. Open cavities. A rectangular open cavity embedded in a wall on which a flat-plate boundary layer develops as a consequence of steady free-stream flow is defined by two independent Reynolds numbers: Re_D , based on the free-stream velocity and a cavity dimension, here the cavity depth, and Re_θ , based on an integral quantity of the incoming boundary layer at a characteristic location at the wall, say momentum thickness, θ , at the upstream cavity lip. Compressibility adds Mach number as a third dimension in the parameter space. All three parameters have been taken into account in the compressible 3D ($\beta \neq 0$) modal BiGlobal instability analysis of Bres & Colonius (2008), in which stability boundaries were defined as a function of these parameters.

The shallow incompressible open cavity analyzed with respect to its 2D ($\beta = 0$) BiGlobal instability by Åkervik et al. (2007) is also embedded in a flat-plate boundary layer. However, its dimension is comparable with the boundary-layer thickness, and the lips of the cavity are rounded, making quantitative comparisons impossible between the results in this geometry and those in the incompressible 2D limit of Bres & Colonius (2008). However, Åkervik et al. (2007) presented the first nonmodal BiGlobal analysis of an open cavity flow, in which optimal perturbations were recovered and utilized to build a control strategy for flow stabilization.

Finally, Sipp & Lebedev (2007) solved the 2D incompressible BiGlobal EVP in a cavity geometry akin to that of Bres & Colonius (2008), employing rather different boundary conditions. The upper boundary of the domain considered in the former work is placed at a height equal to half the cavity depth. At that boundary, a reflection boundary condition is imposed, which alters the downstream development of the boundary layer (and its associated modes) compared with those in the open flow. This also prohibits comparisons with the results of Bres & Colonius (2008) in the incompressible limit.

In all, the multiparametric nature of the open cavity problem, in conjunction with the certainly unintentional but somewhat unfortunate choice of different parameters and boundary conditions in three independent analyses, leaves a wide space of unexplored conditions in this important flow. Experiments focusing on (global) instability mechanisms in a (well-defined) open cavity are highly desirable.

3.3. Leading-Edge Flows

Continuing with open systems, here we discuss leading-edge flows, global instability analyses of which have been performed continuously over the past 15 years, making this one of the most successful applications of global linear instability theory. Work up to the discovery of the polynomial model³ for swept Hiemenz flow (Theofilis 1997, Theofilis et al. 2003) has been reviewed in Theofilis (2003).

Two contributions of Obrist & Schmid (2003) have provided an alternative description of the eigenspectrum of this flow, based on Hermite polynomials. The first nonmodal analysis of leading-edge flows was also performed by these authors, who demonstrated the role played by the continuous spectrum in increasing transient growth. The adjoint BiGlobal equations were derived and used to perform receptivity studies, showing the conversion of free-stream vortical

³The polynomial model has reduced the cost of performing global analysis from $O(h)$ in the spatial direct numerical simulation work of Joslin (1996) or the spatial BiGlobal analysis of Heeg & Geurts (1998) to $O(s)$, without loss of physical information in the linear regime.

disturbances into the leading global eigenmode. Optimal perturbations and optimal control of the swept Hiemenz flow were discussed in a series of publications by Guégan et al. (2008). From a numerical point of view, the latter work is akin to that of Galletti & Bottaro (2004) in that it is based on a set of evolution equations based on the PSE-3D concept, i.e., equations that are parabolized along one spatial direction (here the spanwise direction), which has been treated as periodic in all previous (and subsequent) analyses of leading-edge flows. Still in the incompressible regime, Carpenter et al. (2010) presented modal direct (instability) and adjoint (receptivity) analyses of the swept Hiemenz boundary layer. The demonstration of reliable buffer-domain-based outflow boundary conditions and the feasibility of high-resolution analyses for engineering predictions in the vicinity of the leading edge are both instrumental in the ongoing efforts of this group to provide theoretical support for the associated flight experimentation work.

The first compressible global instability analysis of swept leading-edge flow was performed by Theofilis et al. (2006), who presented BiGlobal EVP and asymptotic analysis results along the lines discussed in their earlier incompressible work. The polynomial model could be extended to compressible flow and could deliver quantitatively correct predictions at low and high subsonic Mach numbers. Li & Choudhari (2008) presented a spatial compressible BiGlobal instability solver, applied to flow in the vicinity of a swept leading edge. A significant finding is that a broad spectrum of stationary cross-flow modes can provoke strong amplification of secondary instabilities. Consequently, early transition can be expected, a finding that questions current transition prediction criteria that are based on primary modal instability results. Finally, Mack & Schmid (2010) also recently addressed the global instability of compressible swept leading-edge boundary-layer flow. Unlike the boundary-layer basic state used by Theofilis et al. (2006), these authors implemented time-stepping tools into a well-validated direct numerical simulation code (Sesterhenn 2001), first to obtain the basic state and then to analyze instability both in the vicinity of the attachment line and in an appreciable portion of the flow away from it, in a region where cross-flow instability dominates. The main result from a physical viewpoint is that the polynomial and the cross-flow modes not only are connected, as predicted by Bertolotti (2000) in the incompressible limit, but can actually be viewed as part of the same amplitude function. Consequently, the long-standing dichotomy between attachment-line and cross-flow modes (Poll 1978) may now be overcome, and the global instability of swept leading-edge flows, at least in a primary linear modal context, may be treated as one physical mechanism.

3.4. Laminar Separation Bubbles

A major breakthrough in the understanding of instability mechanisms of laminar separation bubbles is offered by global linear theory, which considers a 2D basic flow, $\bar{\mathbf{q}} = \bar{\mathbf{q}}(x, y)$, comprising the entire boundary layer in the vicinity of an embedded laminar separation bubble. Amplification of self-excited, as well as incoming, disturbances may then be examined in a unified framework.

3.4.1. The flat-plate boundary layer and two backward-facing geometries. An extensive discussion of the first works devoted to global instability analysis in laminar separation bubbles may be found in Theofilis (2003). Briefly, both absolute/convective theory (Hammond & Redekopp 1998) and the solution of the pertinent BiGlobal EVP (Theofilis et al. 2000) have conclusively demonstrated the potential of this flow to support self-excited global modes, besides the well-known amplification of incoming 2D and 3D disturbances. Hammond & Redekopp (1998) discussed 2D perturbations, whereas Theofilis et al. (2000) quantified the unstable global mode as a stationary 3D perturbation. Independently, the latter conclusion was reached, also via BiGlobal analysis,

by Barkley et al. (2002) on the laminar separation bubble associated with low-Reynolds number steady 2D flow in a backward-facing step.

However, quantitative differences exist in the predictions of the two global instability theories, at least if the magnitude of the peak-recirculation velocity of the basic flow bubble were to be used as a criterion for the onset of self-excitation. The model of Hammond & Redekopp (1998) predicts a level of $O(20\%–30\%)$ for this quantity, whereas both Theofilis et al. (2000) and recent work by Rodríguez & Theofilis (2010b) found that levels of $O(10\%)$ or lower are sufficient for the instability of the 3D stationary global mode. This discrepancy may point to different physical mechanisms at play. Indeed, Rist & Maucher (2002) refined the predictions of Hammond & Redekopp (1998), distinguishing between the viscous and inviscid nature of potentially absolutely unstable 2D eigenmodes, which are fundamentally different from the 3D stationary global mode, and went on to embed this global amplification scenario in a feedback mechanism.

Alternatively, several researchers have confirmed the existence of the self-excited 3D stationary global mode: Merle et al. (2010) in a flow configuration directly comparable with that analyzed by Theofilis et al. (2000), Gallaire et al. (2007) in their analysis of the laminar separation bubble generated by a mild protrusion on a flat surface, and Marquet et al. (2008) in the modal part of their analysis of flow in an S-shaped duct, which is topologically equivalent to the geometry analyzed by Barkley et al. (2002). Experimental evidence of this eigenmode was independently provided by Beaudoin et al. (2004), who observed the appearance of a steady transverse-periodic structure in a backward-facing step flow, whereas Jacobs & Bragg (2006) attributed observations made on iced airfoils to instability having its origins in the 3D stationary global mode.

Consistently strong transient growth of optimal (linear, nonmodal) perturbations has been discovered with regard to nonmodal BiGlobal instability analyses of four different laminar separated flow configurations: an S-shaped rounded-corner duct geometry (Marquet et al. 2008), a geometry-induced separation on a flat plate (Ehrenstein & Gallaire 2008) a backward-facing step (Blackburn et al. 2008a, Barkley et al. 2008), and the aforementioned rounded-corner open cavity (Åkervik et al. 2007). In the S-shaped duct and backward-facing step geometries, the optimal perturbations assume the form of structures filling the entire (duct) geometry and traveling along the streamwise spatial direction. Both the strong growth and the form of the optimal disturbances contrast sharply with the feeble amplification and localized spatial structure of the amplitude functions of the stationary 3D modal perturbation, underlining the distinct origin of the two global linear mechanisms. In addition, the nonmodal scenario clarified the long-standing discrepancy between the stationary 3D global mode predictions and the different observations in the high-fidelity direct numerical simulations of Kaiktsis et al. (1996). It is now accepted that the unsteadiness observed in the backstep simulations is the result of nonmodal instability.

3.4.2. Airfoils and low-pressure turbine blades. The time-stepping modal BiGlobal analysis of Theofilis et al. (2002) monitored incompressible flow around a NACA-0012 airfoil at a small angle of incidence and recovered instability in the wake as the leading global eigenmode of the steady 2D flow at $Re = 10^3$. Kitsios et al. (2009) analyzed flow around a stalled NACA-0015 airfoil at a Reynolds number that was one order of magnitude smaller and identified the same traveling wake mode and, in addition, the stationary 3D global mode as distinct eigenspectrum branches. Abdessemed et al. (2009) found the same two branches in the modal primary analysis of steady laminar flow around a periodic cascade of low-pressure turbine blades. These authors also performed secondary instability analysis, employing Floquet theory, and pseudospectrum studies, whereas the full transient-growth analysis in the cascade was accomplished by Sharma et al. (2010). As anticipated by R. Rivir (private communication, 2006), the transient growth mechanism was found to be the strongest of the three linear instability scenarios investigated in the cascade.

In a series of 3D direct numerical simulation studies of the NACA-0012 airfoil, also at a small angle of incidence but at a chord Reynolds number $Re = 10^4$, Sandham and coworkers explored the connection of global instability in laminar separation bubbles (with laminar separation and turbulent reattachment in a 3D context implied) with self-sustained turbulence. Absolute instability analysis of the flow by both Jones et al. (2008) and earlier work by the same group could not identify the large level of recirculation found by Hammond & Redekopp (1998) to be necessary for global instability. This led Jones et al. (2008) to suggest that the vortex shedding observed in the simulations might be associated with the global mechanism proposed by Theofilis et al. (2000). The spanwise regularity of the streamwise vorticity contours extracted from the simulation is certainly consistent with the 3D nature of the stationary global mode. At this point, further work is necessary in this challenging and technologically significant area.

3.4.3. Separation in transonic and supersonic flow. In their pioneering work, Eriksson & Rizzi (1985) analyzed the global instability of inviscid transonic flow also over the NACA-0012 airfoil. However, the intrinsically viscous phenomenon of shock/boundary-layer interaction was first analyzed by global instability theory by Crouch et al. (2007). These authors monitored turbulent transonic flow over the NACA-0012 airfoil at an angle of attack and flight Reynolds numbers. They were the first to employ global theory to a mean turbulent flow in which an engineering turbulence model was utilized. The key result of Crouch et al. (2007) has been the conclusive demonstration, cross-verified by experiment, that buffeting on an airfoil has its origins in the linear amplification of the low-frequency leading global eigenmode of the entire airfoil/shock system. Given that their analysis is 2D, so is the global mode responsible for buffeting.

Touber & Sandham (2009) analyzed a mean turbulent separation bubble generated by shock/boundary-layer interaction. The flow was resolved by large-eddy simulation, which produced mean results in excellent agreement with experiment. As the global analysis was 3D, it was possible to verify that the most amplified global mode, which is responsible for the observed low-frequency shock motion, is 2D in nature. This is in line with the predictions of Crouch et al. (2007) and is unlike previously reported results in laminar flow by Robinet (2007), providing motivation for further work.

Finally, Sandberg & Fasel (2006) employed direct linear and nonlinear numerical simulations of supersonic axisymmetric wakes in an attempt to reconcile the instability structures associated with the large-scale separation with the predictions of the local theory, which they also performed. They presented evidence regarding the possibility of the coexistence of convectively and absolutely unstable global modes. However, global analysis along the lines discussed by Sanmiguel-Rojas et al. (2009) in the incompressible limit has not been performed in supersonic axisymmetric wake flows.

3.4.4. On the topology of globally unstable laminar separation bubbles. Recent work has linked linear amplification of the unstable stationary 3D global eigenmode of laminar separation bubble to patterns observed in classical topological analyses of 3D separated flows (Dallmann 1982, Hornung & Perry 1984). The seeds of this idea were first articulated by Theofilis et al. (2000), who presented qualitative topological descriptions of shedding from a globally unstable 2D laminar separation bubble, as well as 3D wall-streamline modifications expected on account of the global instability of the entire recirculation zone. Referring to work by Gölling (2001) on a circular cylinder and Schewe (2001) on an airfoil, Theofilis (2003) speculated regarding surface streamline topology:

It has been conjectured . . . that the interaction of the laminar boundary layer at the cylinder surface with the unsteady flowfield in the near-wake is the reason for the appearance (through a BiGlobal instability

mechanism) of the large-scale spanwise periodic structures on the cylinder surface itself. Whether this mechanism can be described in the framework of a BiGlobal linear instability of separated flow or bluff-body instability deserves further investigation.

Such investigation was undertaken recently in two geometries, employing critical-point theory to composite flow fields constructed by linear superposition of the stationary 3D global eigenmode developing upon the laminar separation bubbles formed in an adverse-pressure-gradient boundary-layer flow on a flat plate and a stalled NACA-0015 airfoil. Flow topology theory (Perry & Chong 1987) was used to classify critical points and analyze the bifurcations arising from linear amplification of the global eigenmode in the two geometries.

Focusing on field critical points in a flat plate, Rodríguez & Theofilis (2010b) showed that striking similarities exist between the topology of the composite 3D field and the well-known U-separation pattern (Hornung & Perry 1984). Both are presented in **Figures 4a,b**. In an analogous manner, Rodríguez & Theofilis (2010a) focused on wall streamlines on the airfoil surface (**Figure 4c**). It was shown that linear amplification of the stationary 3D global mode gives rise to a pattern akin to the stall cells seen in a multitude of wind-tunnel experiments and in flight (H. Fasel, personal communication, 2008).

3.5. Bluff Bodies

Instability analysis of bluff-body flows requires, in principle, the application of global modal or nonmodal tools. At the same time, absolute/convective analysis in selected locations of bluff-body wakes may identify global instability in the sense of Huerre & Monkewitz (1990), thus permitting comparisons between the results of the respective theories.

3.5.1. Cylinders of circular and elliptic cross section. The early successful works associated with instability in the wake of a circular cylinder have been sufficiently described in the literature. It suffices here to acknowledge those works as precursors of renewed interest in flow sensitivity and theoretically founded flow control (Giannetti & Luchini 2007).

3.5.2. From the sphere to the prolate spheroid. The prototype geometry on which TriGlobal flow instability may be studied, namely flow over a sphere, is one that permits a reduction of the TriGlobal to a sequence of axisymmetric and nonaxisymmetric BiGlobal EVPs. The early BiGlobal EVP results of Kim & Pearlstein (1990) have been corrected independently in an analysis by Natarajan & Acrivos (1993) and a simulation by Tomboulides et al. (1993). The predictions of the last two works, of a regular bifurcation at $Re \approx 212$ followed by a Hopf bifurcation at $Re \approx 272$, are now the accepted results of global instability in this problem. Subsequent work by Tomboulides & Orszag (2000) elaborated on the instability physics, whereas BiGlobal analyses by Ghidersa & Dusek (2000) and Pier (2008) both confirmed these predictions. Furthermore, Ghidersa & Dusek (2000) discussed axisymmetry breaking from a nonlinear theoretical viewpoint and explained the appearance of the double-thread wake as the superposition of the most unstable global mode to the axisymmetric flow field. Finally, Pier (2008) identified regions of local absolute instability in the wake and presented them as evidence for the consistency in the predictions of absolute/convective instability theory and results of the global EVP.

The first ever TriGlobal analysis to appear in the literature, namely the work of Tezuka & Suzuki (2006), utilized the sphere results for validation of its algorithm. Subsequently these authors computed the first global modes of a nonaxisymmetric 3D object, a prolate spheroid at an angle of incidence. In a manner analogous to the sphere, depending on the angle of incidence, the flow was

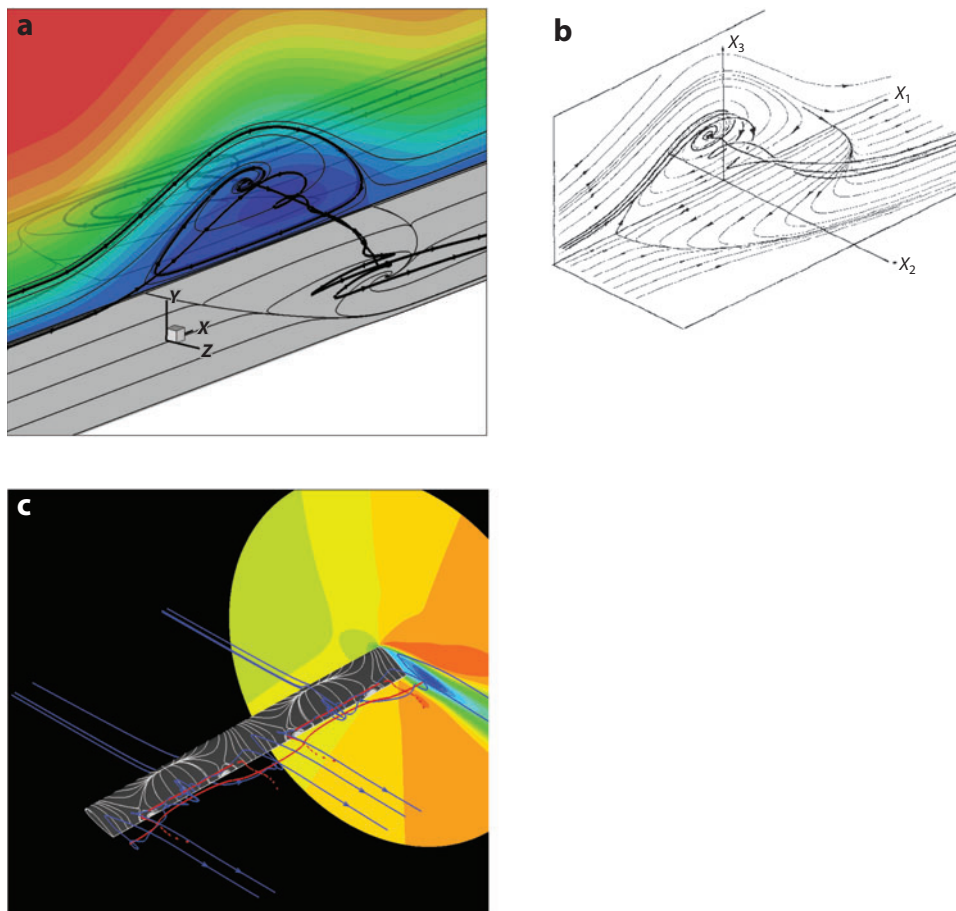


Figure 4

(a) U separation, 2010. Figure taken from Rodríguez & Theofilis (2010b). (b) U separation, 1987. Figure taken from Perry & Chong (1987). (c) Stall cells on the surface of an airfoil. Shown on the perpendicular plane is the spatial distribution of the dominant streamwise velocity component of the leading eigenmode. Figure taken from Rodríguez & Theofilis (2010a).

found to undergo successive bifurcations to steady axisymmetric and nonaxisymmetric patterns as the Reynolds number increases, before unsteadiness finally sets in. The results were found to be in broad agreement with the experimental study also undertaken in that work.

Morzyński & Thiele (2008) also utilized a sphere to demonstrate their TriGlobal analysis algorithm and presented instability results at $Re = 300$. Of interest is the validation work on the Stokes eigenmodes of a 3D, spanwise-homogeneous cylinder. Contrary to expectation, these modes were found to be inhomogeneous along the spanwise spatial direction. This result could only be recovered by TriGlobal analysis, as spanwise homogeneity would be imposed in a BiGlobal analysis context.

Before closing this section, we note that the situation with regard to 3D global instability analyses is at present analogous with that in which Noack & Eckelmann (1994) produced their classic work on reduced-order models for the analysis of 2D global instability in the wake of

a circular cylinder. Reduced-order models for efficient analysis of 3D flows are presently both computationally feasible and physically desirable.

3.6. Systems of Vortices

It is rather surprising that, although the first application of global instability analysis was on systems of vortices (Pierrehumbert & Widnall 1982), it was almost two decades before such analysis came again to the fore. Reasons for this delay include the capability of the simple vortex filament method to predict both modal and nonmodal instability of vortex systems (Crouch 1997) and the successful recovery of elliptical instability physics (Kerswell 2002) through consideration of the axisymmetric flow of one (infinitely long, parallel) vortex being modified by a linear strain accounting for the presence of additional (model) vortices.

The introduction of the Airbus A380 motivated a large number of European projects, FAR-Wake being the latest (<http://www.far-wake.org>), aimed at elucidating the structure and stability of systems of trailing vortices behind commercial airliners. Initial efforts have been reviewed in Theofilis (2003). More recently, following the successful identification of a transient growth scenario on an isolated model vortex by Antkowiak & Brancher (2007), nonmodal work studying global optimal perturbations in vortex systems has been performed. Brion et al. (2007) demonstrated that exciting the vortex dipole with the adjoint eigenmode corresponding to the most unstable (Crow) modal perturbation leads to an energy growth of two to three orders of magnitude for the leading modal perturbation and, potentially, to its rapid breakdown.

The invariably employed assumption of axial homogeneity of a basic flow vortex (system) in all previous global analyses was relaxed by Broadhurst & Sherwin (2008) and Heaton et al. (2009), both of whom independently studied the more-realistic spatially developing Batchelor vortex model. The latter work documented the modifications that the classic Batchelor eigenspectrum experiences and presented nonmodal analysis and the ensuing potential for transient growth exhibited by the spatially developing flow. Direct numerical simulation results in the earlier work of Broadhurst & Sherwin (2008) showed that the nonlinear development of instability in the Batchelor vortex is qualitatively different if the axial direction is kept homogeneous or the vortex is allowed to evolve. In the first case nonlinear saturation is predicted, whereas in the second case vortex breakdown occurs. Motivated by this finding, these authors have invoked a multiple-scales argument and derived an extension of the classic PSE, denoted PSE-3D, to study the instability of more realistic, spatially developing vortex systems. Their analysis is conceptually related to those by Galletti & Bottaro (2004) and Guégan et al. (2008) and is one of the interesting paths that global instability theory may take in the near future.

Finally, Duck (2010) presented inviscid global instability analysis following a rational high-Reynolds number multiple-scales approach, in which the slow (long-wavelength) development of the basic state is also distinguished from the fast (short-wavelength) length scale on which the perturbations evolve. The 2D Rayleigh equation was solved, and its predictions were found to compare favorably with those of independently performed direct numerical simulations. The effect of a pressure gradient was found to be in line with the classic predictions of Hall (1972) on vortex breakdown. The analysis of realistic, nonaxisymmetric, axially inhomogeneous vortex systems is a field in which global instability analysis may provide useful contributions in the near future.

3.7. Non-Newtonian and Physiological Flows

Non-Newtonian and physiological-type flows (grouped together, despite the diverse fields of the associated applications) are the least explored flows by global instability analysis and, at the

same time, most important from an application point of view. Fiétier (2003) presented the first global instability analysis of a viscoelastic flow problem in a complex duct geometry, in which the large degree of non-normality of the associated linear operators was demonstrated through the overlapping of eigenvectors identified in the analysis.

The pioneering work of Sherwin & Blackburn (2005) on pulsatile flow instability in straight tubes with a smooth axisymmetric constriction has been motivated by the need to understand the relation between atherosclerotic plaque formation and pulsatile separated flow instability. Floquet theory was applied to analyze the time-periodic pulsatile flow. Three classes of global modal linear perturbations were discovered: one associated with ring formation at high velocities, another taking the form of wave-like perturbations at lower speeds, and, under certain conditions, convective instability and localized turbulence generation at the separated shear layer past the stenosis. The follow-up study by Blackburn et al. (2008b) addressed nonmodal instability in the same geometry and demonstrated the large transient growth associated with the latter of the three mechanisms, namely localized convective instability in extended shear layers.

This nonmodal mechanism was first predicted for weakly nonparallel flow by Cossu & Chomaz (1997), is analogous in nature with that discovered in other separated flows, and provides a fitting closure of the present review by serving as a reminder that knowledge acquired through simpler approaches may still be useful in interpreting results arising in open flows over or through more complex geometries.

4. FINAL REMARKS

The present review has intended to build a bridge between the singular achievements of three decades ago and the present-day routine performance of modal and nonmodal global (BiGlobal) instability analyses in flows with two inhomogeneous and one homogeneous spatial directions. The 3D (TriGlobal) analyses already available in the literature demonstrate that no fundamental obstacles exist in the extension of these ideas and tools to the new frontier of 3D flow instability. Above we attempt to demystify these analysis tools to raise the interest of current and future global instability practitioners in contributing to the development of next-generation algorithms exploiting present-day computing hardware and enabling further progress.

Results reviewed demonstrate that algorithms and hardware have matured sufficiently to address the instability of flows beyond the restrictive assumption of a weakly nonparallel basic flow. Global modal and nonmodal analysis based on solution of multidimensional EVPs and IVPs presently not only routinely predicts critical conditions for the instability of laminar flows in complex geometries, but also underpins theoretically founded methodologies for the identification of regions of sensitivity and control of such flows (Giannetti & Luchini 2007), provides answers to industrially relevant turbulent flow problems (Crouch et al. 2007), and forms an essential part of successful reduced modeling efforts (Noack et al. 2003).

However, global instability experimentation presently lags well behind the fast pace at which theoretical results fill the pages of high-Impact-Factor journals. Experiments validating theoretical results are urgently needed to demonstrate the relevance of the analysis to practical situations, most of which concern flows developing in complex 2D and 3D geometries. The combination of the advanced theoretical/numerical tools of global instability analysis with experimental verification of theoretical predictions is the safest means of not only preventing the analysis from becoming an interesting academic exercise but, more importantly, permitting it to reach its full potential as an enabling technology for the solution of real-world problems.

SUMMARY POINTS

1. If a 1D basic flow is to be analyzed, classic linear theory leading to EVP of the Orr-Sommerfeld class or to the equivalent IVP is the most accurate and efficient theoretical analysis tool. For weakly 2D flows, PSE (Herbert 1997), which can be straightforwardly extended to perform nonlinear instability analysis, should be used.
2. Predictions of global instability may be based on the theories discussed by Chomaz et al. (1988) and Huerre & Monkewitz (1990), as long as the weakly nonparallel flow assumption is satisfied along one spatial direction in a given 2D or 3D application.
3. The global instability analysis methodology discussed above extends the theories underpinning points 1 and 2 above in 3D flows with two or three inhomogeneous spatial directions. IVP-based nonmodal analysis is an essential complement of EVP solutions.
4. BiGlobal modal and nonmodal analysis of laminar or turbulent flows is feasible on a routine basis using present-day technology and algorithms.
5. Both turbulence and shocks have been successfully modeled in industrial flows with two inhomogeneous and one homogeneous spatial directions, paving the way for the same to be accomplished in fully 3D flows.
6. Receptivity, sensitivity, and, ultimately, control of flows in complex domains may be performed, from first principles, using theoretically founded methodologies, at no extra cost to that of the instability analysis.
7. Reduced-order models may improve their performance by including global instability analysis results.
8. The quality of instability analysis results in complex geometries is independent of the spatial discretization method employed. However, next-generation eigenspectrum computation algorithms, taking advantage of present-day technology, would accelerate progress in both the BiGlobal and TriGlobal frontiers.

FUTURE ISSUES

1. Work on inflow and outflow boundary conditions for open problems is needed, especially in applications involving (hydrodynamic or aeroacoustic) wave propagation.
2. Algorithms that circumvent matrix storage and inversion hold promise to provide the breakthrough necessary for TriGlobal flow instability analysis (and control).
3. Reduced-order models for complex flows, when available, are expected to drastically reduce the cost of performing global instability analysis.
4. Treatment of turbulence in predictive instability analysis and control models needs to be addressed, focusing on complex geometries.

NOTE ADDED IN PROOF

Another modal TriGlobal analysis of cubic lid-driven cavity flow was presented by Feldman & Gelfgat (2010), as this article was going into print.

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See Supplemental Appendix 2 to obtain a copy of this paper.

Presents a global analysis of turbulent transonic flow; identifies global instability of turbulent shock/boundary-layer interaction at the origin of buffeting.

First work to introduce critical-point theory to the topological description of 3D flows.

Introduces time-stepping and spectral transformation to the stability analysis of inhomogeneous steady and time-periodic flows.

Rediscovered, extended, and popularized theoretically founded receptivity and sensitivity analysis.

Introduces adjoint analysis to control of complex flows.

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First global analysis of 3D flow, reduced to the solution of 2D EVPs by exploitation of symmetries.

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